Shane Sweetman

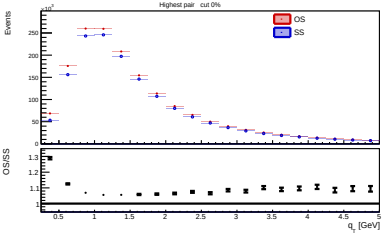
Project update meeting
Week 8

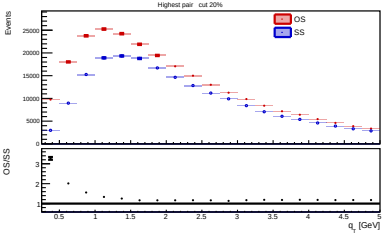
A TMD-oriented analysis of $\pi^+\pi^-$ pairs in e^+e^- collisions

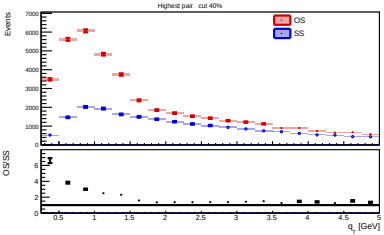
Outline (Week 8)

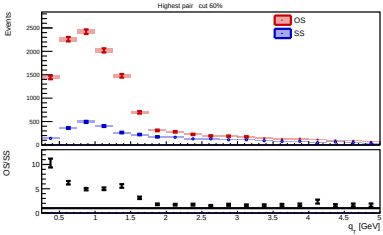
- Final OS vs SS distributions.
- OS/SS ratio as a function of the pion-fraction cut.
- Fit comparison plot.
- Chi-square comparison across cuts.

OS vs SS distributions

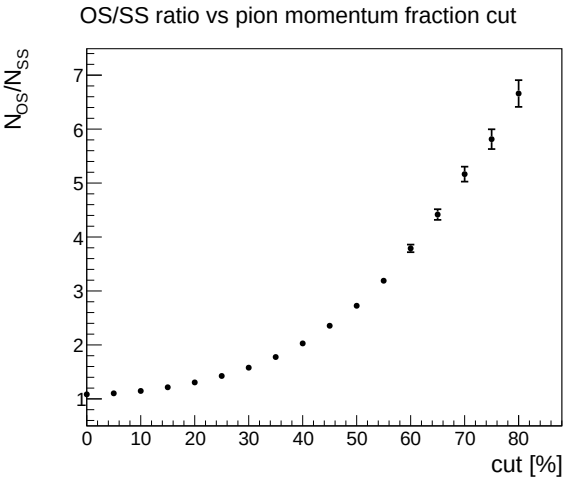




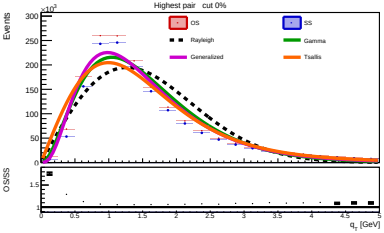


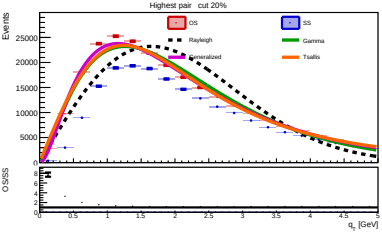


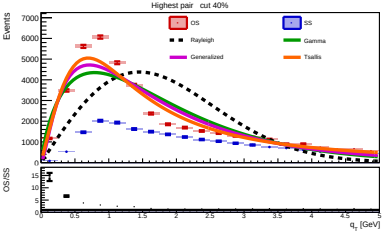
OS/SS ratio vs pion-fraction cut

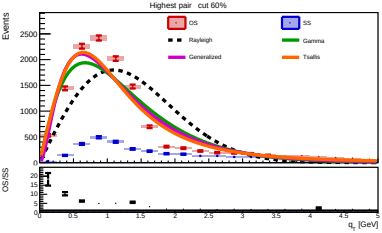


Fit comparison









Definitions of the four fit forms

Rayleigh

$$f(q_T) = A\, q_T\, \exp\left(-\frac{q_T^2}{2\sigma^2}\right)$$

Generalized / Weibull

$$f(q_T) = A\, q_T^{\alpha} \exp\left[-\left(\frac{q_T}{\lambda}\right)^m\right]$$

Gamma / exponential-tail

$$f(q_T) = A\, q_T^{\alpha} \exp(-\beta q_T)$$

Tsallis / power-law tail

$$f(q_T) = A\, q_T \left(1 + \frac{q_T^2}{Q_0^2}\right)^{-p}$$

Chi-square comparison of fit forms

Rayleigh

Cut	χ^2	DoF	χ^2/DoF
0%	234238	18	13013.2
20%	14076.0	18	781.998
40%	14318.8	18	795.491
60%	5689.04	18	316.058

Gamma

Cut	χ^2	DoF	χ^2/DoF
0%	77072.4	17	4533.67
20%	1632.11	17	96.0067
40%	3416.50	17	200.971
60%	1850.60	17	108.859

Generalized / Weibull

Cut	χ^2	DoF	χ^2/DoF
0%	45600	16	2850.00
20%	722.469	16	45.1543
40%	2103.66	16	131.479
60%	1350.67	16	84.4169

Tsallis

Cut	χ^2	DoF	χ^2/DoF
0%	107479	17	6322.32
20%	1411.80	17	83.0471
40%	1199.78	17	70.5751
60%	997.970	17	58.7041

Best reduced χ^2: Generalized at 0% and 20%, Tsallis at 40% and 60%.
Overall very bad fits and not physics motivated.

Questions?

References for fit forms

[1] M. M. Siddiqui, *Statistical Inference for Rayleigh Distributions*, Journal of Research of the National Bureau of Standards **68D** (1964).

[2] NIST/SEMATECH e-Handbook of Statistical Methods, *Gamma Distribution*.

[3] E. W. Stacy, *A Generalization of the Gamma Distribution*, Annals of Mathematical Statistics **33** (1962) 1187–1192.

[4] M. Boglione, J. O. Gonzalez-Hernandez, and R. Taghavi, *Transverse parton momenta in single inclusive hadron production in e^+e^- annihilation processes*, Physics Letters B **772** (2017) 78–86.